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Bearing Damage Analysis - Methodology

Presented at the bearing damage course, WE 204

Presented by xxx

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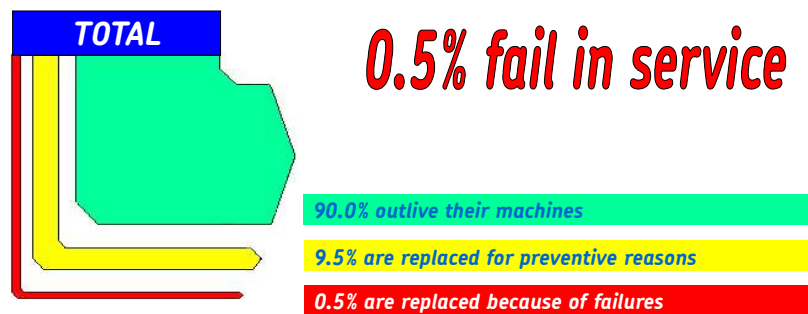
Bearing failures

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Some bearings fail during service



- *In industrial applications, the number can be higher*
- *Most root causes are a result of maintenance and installation issues*

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Bearing failure causes



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Root cause failure analysis

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Bearing damage analysis...

When a bearing fails, the evidence of the events that led to the failure most of the time does show up in the bearing.

You are now placed in the role of a bearing detective.



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Bearing damage analysis...

- *Where, when, and why did the failure occur?*
- *One key piece of forensic evidence is the bearing itself*
- *A variety of causes lead to bearing damage and ultimately failure*
- *Causes, such as improper installation, inadequate lubrication, contamination, and improper operation, leave distinctive marks on the bearing, be it on the raceways, cage(s), rolling elements, bore, outside diameter, or others*

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Bearing damage analysis



Noting and interpreting these readings can provide a remarkably accurate assessment of the failure mode, and more importantly, lead to identifying corrective actions to reduce the incidence of future failures.

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Definition

- **Root Cause Failure Analysis is a method to determine the primary cause of a failure or a problem, and to eliminate it**
- **It is finding the true cause of the problem instead of focusing on the apparent cause**



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An example

- **Problem:**
 - *Your car's engine overheats and shuts down*
 - *What caused the engine to overheat?*
- **Observation:**
 - *The water pump belts were loose*
- **Solution**
 - *Tighten the belts*



WRONG!

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Why was it wrong ?

- *Only a partial analysis was carried out*
- *Why were the belts loose?*
- *Leaves a returning problem...*

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Using RCFA – observations

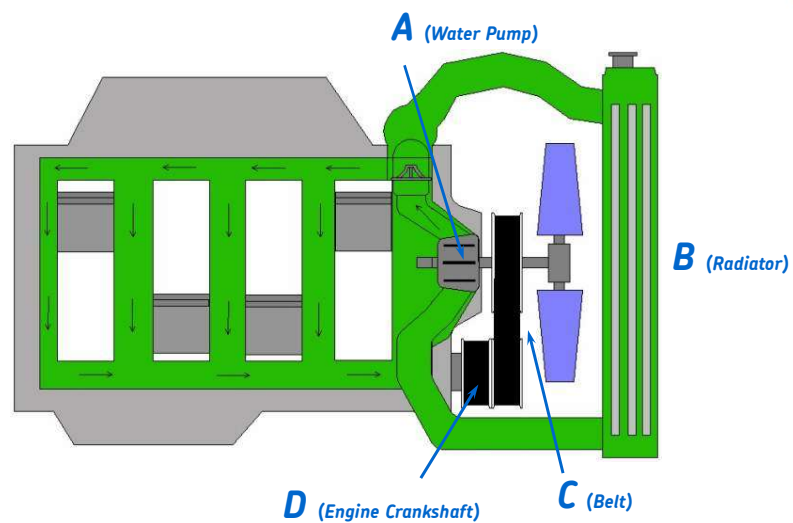
- *Engine overheated and shut down*
- *Water pump belts loose*
- *Belts in good physical condition...*

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Using RCFA – decomposition



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Using RCFA – cause of overheating

- *The belts are not tight enough*
- *Why?*

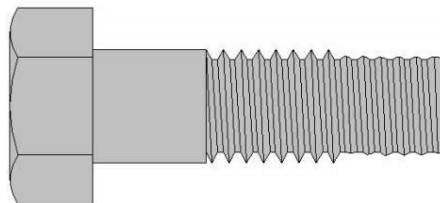
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Using RCFA – the true cause of problems

- *Worn bolt thread*
- *Replace the tensioning bolt, AND*
- *Verify all procedures and operating conditions*



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Why, why, why, why, why?

<i>Why was the bolt stripped?</i>	<i>Over-tightened</i>
<i>Why was it over-tightened?</i>	<i>Mechanic did not use torque wrench</i>
<i>Why no torque wrench?</i>	<i>Not trained properly</i>
<i>Why not trained?</i>	<i>Time and money</i>

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Bearing analysis preparation

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Bearing analysis – preparation

- *In situ inspection (Before dismounting)*
- *Dismounting and component conservation*
- *Information gathering by the user*
- *Information gathering by the analyst*

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In situ inspection

- *Position of the bearing in the machine*
- *Details about the failure (e.g., lubricant condition, contamination, shaft and housing condition, frame)*
- *Position of the broken parts*
- *Actual radial and axial clearance*
- *Consequences of the failure, secondary damages, etc.*

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Dismounting and bearing protection

- **SKF dismounting guidelines** → (refer to mounting/dismounting tab of individual product)
- **Take (digital) pictures before, during, and after dismounting!**
- **Prevent or minimize further damage**
- **Protect from further deterioration such as corrosion**

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Information gathering by the user

- **Machine and bearing history**
- **Application drawing(s)**
- **Maintenance history and follow-up**
- **Vibration analysis readings (if available)**
- **Problem report**

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Information gathering by the analyst (1)

- *Complete bearing designation(s)*
- *Type of machine*
- *Mounting techniques used*
- *Rotational speed*
- *Axial and radial loads*
- *Bearing operating temperature*
- *Ambient temperature and humidity*

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Information gathering by the analyst (2)

- *Environment, contamination level*
- *Operating time (24 hrs/day, 7days/week)*
- *Lubricant used and its specifications*
- *Lubrication specifications (method, intervals, quantities, etc.)*
- *Bearing life (calculated, actual, mean time between failures)*
- *Cost due to failure (production loss, parts, labour, etc.)*

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Agreements and arrangements

- *Contact information of the user, manufacturer, supplier*
- *Deadlines of bearing analysis when outsourced*
- *Disposal of the components once the analysis is completed*
- *Format of the final report*

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Bearing analysis

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Taking the bearing apart

- *The SKF dismantling guidelines*
- *For spherical roller bearings*
- *For deep groove ball bearings or double row angular contact ball bearings, contact SKF*
- *For most other bearing types, follow the SKF dismantling guidelines and the bearing falls apart*
- *In general, take (digital) pictures before, during, and after taking the bearing apart!*
- *Mark the components if necessary (to avoid mixing)*

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Preliminary inspection

- *Lubricant condition (colour, presence of water, viscosity, consistency, distribution in the bearing, etc.)*
- *Degree of corrosion*
- *Temperature effects (discolouration, deformation, etc.)*

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Cleaning

- *Remove the excess of grease and clean the bearing components with a solvent*
- *Light surface corrosion and solid grease deposits can carefully be removed with an abrasive pad such as wire wool*

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Visual inspection

- *Inspect each bearing component separately (inner ring, outer ring, cage, rolling elements, seals, etc.)*
- *Take notes on the wear paths observed*
- *Take notes on all damages observed*
- *Identify the damage initiation points*
- *Compare similar components with each other (raceways, rolling elements, inner/outer ring wear patterns)*
- *Eventually, hardness testing of components may be required*
- *Take (digital) pictures of all relevant bearing damage!*

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Lubricant analysis...

- *Take lubricant samples from the sides and the centre of the bearing*
- *Do not contaminate the sample from outside sources!*
- *Condition: colour, consistency, viscosity, presence of water, smell, quantity of grease/oil, etc.*
- *Always compare with a sample of new, unused lubricant*

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Lubricant analysis

- *Use a flat spatula or other device to spread the lubricant out over a piece of clean white paper to see and feel hard materials, ferrous metal particles, or other contaminants*
- *Use a magnifying glass to assist in these observations*
- *Use a small magnet on the underside of the white paper to show if the lubricant is contaminated with metallic particles*

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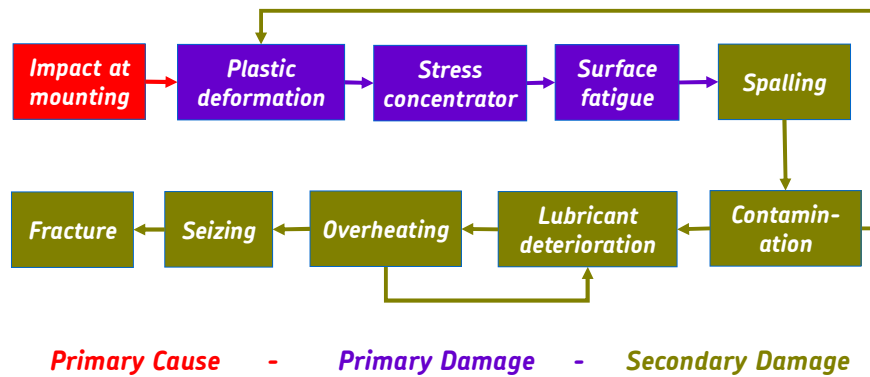
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Notion of primary and secondary damage



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Refrain from preconceived ideas!

- *Blaming things you cannot verify*
- *Making assumptions based on rumours*
- *Personal biases*
- *Outside influences*
- *What else happens at the facility*
- *Inspect the bearing fully*

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Bearing analysis report

An analysis report includes the following sections:

- *Introduction (objectives of the analysis, application history)*
- *Observations (list of damages, photos, etc.)*
- *Failure mode analysis (ISO)*
- *Conclusion (summary of the analysis)*
- *Recommendations*

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Thank you

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